501 Radiation and Radioactivity

5.1.1 Production and properties of X-rays

X-rays are short-wavelength, high-energy electromagnetic waves, produced for example when fast electrons decelerate at a metal target. They need not originate in nuclear decay. Their penetrating and ionising properties, and different attenuation in different materials, allow internal structures to be imaged. Distinguish production, transmission and detection when explaining imaging.

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5.1.2 Nuclear radiation

Unstable nuclei decay spontaneously and emit radiation, commonly alpha particles consisting of helium nuclei, beta-minus particles consisting of fast electrons, and electromagnetic gamma rays. These are nuclear processes, not ordinary outer-electron transitions. Individual decay times are random, although large populations follow predictable average decay laws.

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5.1.3 Ionising ability of nuclear radiation

Ionisation occurs when radiation removes electrons from atoms or molecules, leaving ions. In typical textbook comparisons, alpha radiation ionises most strongly, followed by beta and then gamma. Alpha particles lose energy densely over short paths and therefore penetrate poorly. Strong ionising ability does not imply the greatest range.

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5.1.4 Range of radiation in air

Typical alpha ranges in air are a few centimetres, whereas beta particles may travel a few metres. Actual ranges depend on initial energy and medium. Gamma rays are progressively attenuated rather than having a sharp finite stopping range like charged particles. Use the specified energy, medium and measured graph when provided.

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5.1.5 Random fluctuations in radioactive counts

Radioactive counts fluctuate because decay is random. Measurements made under identical conditions need not be exactly equal, and one high reading does not establish an instrument fault. Subtracting background corrects its contribution but does not remove the intrinsic random fluctuations of the counts.

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5.1.6 Subtracting background counts

A detector records contributions from the source and background. The source's net count rate is measured rate minus background rate, using matching time units. Repeated or longer measurements reduce random fluctuations. Even after background subtraction, the detector usually captures only part of the radiation, so count rate is not automatically total source activity.

$$C_{\text{net}} = C_{\text{measured}} - C_{\text{background}}$$

Corrected count rate subtracts background measured in the same units. Count rate is not automatically the activity because detector efficiency and geometry matter.

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5.1.7 Penetrating ability of nuclear radiation

Typical penetrating ability decreases in the order gamma, beta, alpha. Paper or skin stops alpha particles, a few millimetres of aluminium stops typical beta particles, and thicker lead or concrete substantially attenuates gamma rays. Gamma shielding is normally described by a reduction factor, not complete removal. This ordering differs from ionising ability.

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5.1.8 Properties and detection of nuclear radiation

Alpha particles are helium nuclei of charge +2e, beta-minus particles are fast electrons of charge −e, and gamma rays are uncharged electromagnetic radiation. Alpha and beta bend oppositely in electric fields and experience magnetic force unless moving parallel to the field; gamma is undeflected. Alpha is strongly ionising but stopped by paper, beta is intermediate and typically stopped by millimetres of aluminium, and penetrating gamma needs substantial lead or concrete attenuation. Geiger counters can detect them, provided alpha reaches the sensitive region through a thin window. Cloud chambers show alpha and beta tracks; gamma produces no direct charged-particle track.

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5.1.9 Natural background radiation

Natural background comes mainly from cosmic radiation and radioactive substances on Earth. Textbook examples include radon released from soil and rocks, granite, and trace radioactive material in food and water. A detector can therefore register counts even when no artificial radioactive source is present.

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